Rowan Barbalich

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Education

Stevens Institute of Technology

Master of Mechanical Engineering with Control Systems Concentration 5/20/2026

Related Coursework: Autonomous Navigation for Mobile Robots, Mobile Micro robotic Systems, Modern Control Engineering, Analytical Dynamics

Bachelor of Mechanical Engineering, Minor of Green Engineering 5/2025

Related Coursework: Statics and Introduction to Engineering Mechanics, Circuits and Systems, Thermodynamics, Design of Dynamical Systems, Thermal Engineering, Manufacturing Processes and Systems, Field Sustainable Systems with Sensors, Engineering Economics and Project Management, Fluid Mechanics

Technical Skills

CAD/Simulation: SolidWorks, Blender, MATLAB, ANSYS (Workbench, APDL, Fluent), AutoCAD

Fabrication: CNC Milling, Lathe Operation, TIG Welding, Autodesk Fusion

Programming: MATLAB, C++, Python, Arduino

Other Tools: MS Office Suite, Google Suite, Adobe Photoshop/Illustrator, Blender, AI (Claude, ChatGPT, etc.)

Academic Projects

Programmable Gas Chromatography Oven

- Led the mechanical design team for a programmable gas chromatography oven build. Facilitated weekly progress meetings and enforced scheduling accountability.
- Utilized SolidWorks to create several prototypes and a final design capable of achieving temperature ramp rates up to 400°C in under twenty minutes.
- Implemented and tuned PID temperature control using Arduino, heating elements, and thermocouples, achieving a 30% reduction in settling time once properly tuned.
- Fabricated inner and outer steel enclosures via CNC milling and lathe operation; completed final assembly using riveting and TIG welding.

Autonomous Campus Tour Robot

- Designed and fabricated an autonomous mobile robot in SolidWorks and manufactured the chassis via 3D printing to house LiDAR, ultrasonic sensors, and control electronics
- Integrated YDLIDAR G4 laser scanner for odometry, along with multiple ultrasonic sensors for real-time obstacle detection and collision avoidance in a physical obstacle course.
- Developed a path-planning and target-reaching algorithm in Arduino using live x-y coordinate data streamed over MQTT/Wi-Fi for active navigation.

Mobile Robot Simulation & Closed Loop Control

- Developed a closed-loop control system for a differential-drive mobile robot using ultrasonic sensor feedback for real-time environmental awareness.
- Designed function-based control logic in MATLAB for wall-tracking and obstacle avoidance. Achieved consistent, timely course completion over 5000 simulated trials.

Experience

McGoey, Hauser, and Edsall Consulting Engineers, D.P.C. Engineering

June 2023 – August 2023, Milford PA

- Produced engineering and site plans in AutoCAD for municipal and private infrastructure projects including roadway drainage and structural renovation.
- Conducted field inspections to verify construction compliance with engineering specifications.
- Analyzed and organized watermain material data in Excel for the Town of Port Jervis, NY, supporting infrastructure condition assessment.

Other Experience: SAE Baja, Club Lacrosse Vice President, Engineers without Borders